

Enantioselective Total Synthesis and Antifungal Evaluation of Lycopodine-Derived Alkaloids

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Abstract:

A scalable asymmetric total synthesis of the tetracyclic alkaloid derivative (+)-Lycopodine Analogue 6b has been achieved in 11 linear steps from commercial cyclopentenone. The critical C-C bond forming cascade leverages an intramolecular organocatalytic Michael-aldol cyclization with >99:1 enantiomeric ratio. In vitro biological profiling against pathogenic fungal strains identified Compound 6b as a selective inhibitor of *Candida albicans* biofilm formation, displaying a minimum inhibitory concentration (MIC) of 1.8 $\mu\text{g/mL}$. Deprotection conditions utilizing trifluoroacetic acid (TFA) in dichloromethane at 0 °C afforded the target alkaloid in 84% yield without epimerization.

Introduction:

Lycopodium alkaloids present intricate polycyclic architectures decorated with contiguous stereocenters. Natural isolates frequently exhibit neuroprotective, antimicrobial, and immunosuppressive properties. Accessing non-natural synthetic analogues enables systematic alteration of steric bulk and hydrogen-bonding motifs to dissect structure-activity profiles.

Synthetic Sequence and Asymmetric Construction:

The synthesis initiated with an enantioselective conjugate addition catalyzed by a chiral secondary amine catalyst (10 mol %). The key tricyclic ketone intermediate 5 was assembled through a microwave-assisted cascade sequence in toluene at 110 °C.

Scheme 2 highlights the final deprotection and functionalization:

Intermediate 5b was treated with trifluoroacetic acid (TFA, 20% v/v) in anhydrous CH₂Cl₂ at 0 °C for 45 min under a nitrogen atmosphere. Neutralization with saturated NaHCO₃ followed by extraction with dichloromethane afforded (+)-Lycopodine Analogue 6b in 84% isolated yield.

Physical and Spectroscopic Data:

Compound 6b: colorless viscous oil; [α]_D²⁰ = +64.2 (c 1.0, CHCl₃); isolated yield: 84%.

¹H NMR (500 MHz, CD₃OD): δ 3.42 (m, 1H), 3.12 (dd, J = 12.0, 4.5 Hz, 1H), 2.85 (m, 2H), 2.45–2.30 (m, 3H), 1.95–1.70 (m, 6H), 1.05 (d, J = 6.5 Hz, 3H).

¹³C NMR (125 MHz, CD₃OD): δ 212.4, 60.1, 52.8, 46.5, 43.2, 38.6, 36.1, 30.5, 27.2, 24.8, 19.3.

HRMS (ESI) m/z calculated for C₁₆H₂₆NO [M+H]⁺ 248.2014, found 248.2011.

Antifungal Activity and Cellular Selectivity:

Synthetic analogues 6a–6d were screened against fungal pathogens *Candida albicans* (ATCC 90028) and *Aspergillus fumigatus* (AF293). Amphotericin B served as the positive clinical benchmark.

Table 1: Antifungal Minimum Inhibitory Concentrations (MIC, $\mu\text{g/mL}$)

Alkaloid Analogue	<i>Candida albicans</i> MIC	<i>Aspergillus fumigatus</i> MIC	Cytotoxicity IC ₅₀ (Vero Cells)
Compound 6a	14.5 $\mu\text{g/mL}$	>64 $\mu\text{g/mL}$	>100 μM
Compound 6b	1.8 $\mu\text{g/mL}$	28.4 $\mu\text{g/mL}$	>100 μM
Compound 6c	8.2 $\mu\text{g/mL}$	>64 $\mu\text{g/mL}$	75.2 μM
Amphotericin B	0.5 $\mu\text{g/mL}$	1.0 $\mu\text{g/mL}$	12.5 μM

Compound 6b demonstrated potent, selective inhibition of *C. albicans* with an MIC of 1.8 $\mu\text{g/mL}$ and favorable therapeutic index (>55-fold selectivity over mammalian Vero cells).

Notice on Scope:

The present manuscript focuses exclusively on in vitro antifungal potency and synthetic methodology. In vivo pharmacokinetic evaluation (e.g., rat oral bioavailability or clearance) and single-crystal X-ray crystallographic coordinates were not investigated in this study.